

Correlation of Emotional Trauma, Chronic Stress, and Autoimmune Diseases

By: Michele Serrano

The Correlation Between Unprocessed Emotional Trauma, Chronic Stress, Inflammation, and the Development of Autoimmune Diseases

Introduction

Unprocessed emotional trauma and chronic stress have long been recognized as significant contributors to mental health disorders, but a growing body of scientific evidence now implicates these psychological factors in the dysregulation of immune function and the pathogenesis of autoimmune diseases. Post-traumatic stress disorder (PTSD), a prototypical outcome of unresolved trauma, is associated not only with psychiatric morbidity but also with a heightened risk of chronic physical illnesses, including a spectrum of autoimmune conditions such as rheumatoid arthritis (RA), systemic lupus erythematosus (SLE), multiple sclerosis (MS), and inflammatory bowel disease (IBD)^{[2][4]}. This report synthesizes current scientific understanding of the epidemiological links, biological mechanisms, and psychological factors that connect trauma and chronic stress to immune dysregulation, inflammation, and autoimmunity. It further explores disease-specific associations, the role of early-life adversity, neuroimmune interactions, and the implications for clinical practice and future research.

Overview: Linking Trauma, Chronic Stress, Inflammation, and Autoimmunity

Epidemiological Evidence

Large-scale cohort studies and meta-analyses consistently demonstrate that individuals with PTSD or significant trauma exposure have a statistically significant increased risk of developing autoimmune diseases. For example, a meta-analysis encompassing nearly two million individuals found that PTSD was associated with a 29% increased risk of any autoimmune disease (RR = 1.291, 95% CI: 1.179-1.412), with disease-specific risk ratios ranging from 1.14 for IBD to 1.30 for MS^[1]. The Millennium Cohort Study of US military personnel reported a 58% higher risk of selected autoimmune diseases among those with PTSD, independent of combat experience, prior trauma, or health behaviors such as smoking and alcohol use^[2]. Similar findings have been observed in civilian populations, female nurses, and veterans, underscoring the robustness of this association across diverse demographics^{[6][4]}.

Biological Mechanisms

The mechanistic pathways linking trauma and chronic stress to immune dysregulation and autoimmunity are multifaceted, involving neuroendocrine, autonomic, and immune system interactions. Chronic stress disrupts the hypothalamic-pituitary-adrenal (HPA) axis, leading to glucocorticoid receptor resistance and impaired anti-inflammatory signaling^[8]. This dysregulation fosters a pro-inflammatory milieu characterized by elevated cytokines (e.g., IL-6, TNF- α , CRP), altered immune cell phenotypes (e.g., increased Th17, reduced Treg), and changes in gene expression and epigenetic regulation^{[10][12][13]}. The sympathetic nervous system (SNS) and catecholamines further modulate immune responses, while neuroimmune interactions and gut brain-immune axis disturbances contribute to systemic inflammation and loss of immune tolerance^[15].

Psychological Factors

Unresolved trauma and maladaptive coping styles perpetuate chronic stress responses, maintaining the body in a state of hyperarousal and immune activation. Early-life adversity, such as adverse childhood experiences (ACEs), has a particularly strong and lasting impact on immune regulation, increasing the risk of autoimmune disease in adulthood^[3]. Psychological interventions that address trauma and promote adaptive coping can modulate inflammatory pathways and may reduce autoimmune risk^[16].

Epidemiological Evidence and Cohort Studies

PTSD and Autoimmune Disease Risk

Multiple prospective and retrospective cohort studies have established PTSD as a significant risk factor for autoimmune diseases. In the Millennium Cohort Study, active-duty US military personnel with PTSD had a hazard ratio (HR) of 1.6 (95% CI: 1.2-2.0) for developing any of the selected autoimmune diseases (RA, SLE, MS, IBD) compared to those without PTSD^[2]. Disease specific HRs ranged from 1.4 for RA and SLE to 2.3 for MS, with similar findings in both men and women^[1].

A meta-analysis of eight large cohort studies found statistically significant associations between PTSD and IBD (RR = 1.142), MS (RR = 1.302), SLE, RA, and thyroiditis^[11]. Subgroup analyses indicated that both male and female PTSD patients had elevated risks (RR = 1.266 for males, 1.285 for females), and higher PTSD symptom burden was associated with greater risk (RR = 1.335 for high symptom burden)^[11].

Early-Life Trauma and Adverse Childhood Experiences (ACEs)

The Adverse Childhood Experiences (ACE) Study, tracking over 15,000 adults, found that individuals with two or more ACEs had a 70-100% increased risk of hospitalization for autoimmune conditions compared to those with none^[3]. Each additional ACE point raised the risk of autoimmune hospitalization by 20% in women and 10% in men. Rheumatic diseases

such

as RA and SLE showed the steepest increases, with a 100% greater risk for those with two or more ACEs^[3].

PTSD-Specific Associations

PTSD is not only associated with increased risk of autoimmune disease onset but also with greater disease activity and severity. For example, veterans with MS and comorbid PTSD had higher relapse rates, more MRI lesions, and faster disability progression than those without PTSD^[17]. In SLE, PTSD was associated with at least a twofold increased risk, and in some studies, nearly threefold, particularly among women^[6].

Table: Key Epidemiological Studies Linking Trauma/PTSD to Autoimmune Disease

Study/Population	Autoimmune Diseases	Key Findings (Risk Ratios/HRs)
Millennium Cohort (US Military)	RA, SLE, MS, IBD	HR = 1.6 for any autoimmune disease with PTSD
Nurses' Health Study II (Women)	RA, SLE, Thyroiditis	HR = 1.76 for RA with ≥ 4 PTSD symptoms
Swedish National Registers	41 autoimmune diseases	RR = 1.09-1.49 (varies by disease)
ACE Study (US Adults)	21 autoimmune diseases	70-100% increased risk with ≥ 2 ACEs
Veterans Health Administration	RA, SLE, MS, IBD, Thyroiditis	Elevated risks in PTSD vs. controls
Meta-analysis (8 studies, n~ 2M)	IBD, SLE, RA, MS, Thyroiditis	RR = 1.14-1.30 (disease-specific)

The consistency of these findings across populations, genders, and trauma types underscores the robustness of the trauma-autoimmunity link^{[3][6][4][1]}.

Biological Mechanisms Linking Trauma and Chronic Stress to Immune Dysregulation

HPA Axis Dysregulation and Glucocorticoid Receptor Resistance

The HPA axis is the central neuroendocrine system mediating stress responses. Acute stress triggers the release of corticotropin-releasing hormone (CRH), adrenocorticotrophic hormone

(ACTH), and ultimately cortisol, which exerts anti-inflammatory effects via glucocorticoid receptors (GRs)^[8]. However, chronic stress and trauma lead to impaired HPA axis feedback, reduced cortisol secretion, and glucocorticoid receptor resistance. This results in diminished anti-inflammatory signaling and a shift toward a pro-inflammatory state^{[7][8]}.

In PTSD, cortisol levels are often paradoxically low, and immune cells become less responsive to glucocorticoids, a phenomenon termed glucocorticoid receptor resistance^[12]. This resistance impairs the body's ability to suppress inflammation, allowing pro-inflammatory cytokines to persist and promote autoimmunity^{[12][8]}.

Sympathetic Nervous System, Catecholamines, and Immune Modulation

The sympathetic nervous system (SNS) is activated during stress, releasing catecholamines (adrenaline, noradrenaline) that modulate immune cell function^[8]. Acute SNS activation can enhance innate immunity, but chronic activation leads to immune dysregulation, including impaired T and B cell function, altered cytokine production, and increased inflammation^[8]. Catecholamines can directly stimulate pro-inflammatory cytokine release and influence the balance of immune cell subsets, such as increasing Th17 and reducing Treg cells^[14].

Cytokines and Inflammatory Signaling Pathways

Chronic stress and PTSD are associated with elevated levels of pro-inflammatory cytokines, including IL-6, TNF- α , IL-1 β , and CRP^{[18][12][16]}. The transcription factor NF- κ B, a master regulator of inflammation, is upregulated in PTSD and chronic stress, further amplifying cytokine production^[12]. Over time, the anti-inflammatory effects of cortisol are blunted, and immune cells become hyper-responsive to stress signals, perpetuating a cycle of inflammation^[12].

Table: Key Inflammatory Biomarkers Elevated in PTSD/Chronic Stress

Biomarker	PTSD/Chronic Stress Effect	Clinical Implications
IL-6	↑	Linked to RA, SLE, CVD, depression
TNF- α	↑	Drives joint, gut, CNS inflammation
CRP	↑	Marker of systemic inflammation
IL-1 β	↑	Promotes neuroinflammation
IL-17	↑	Th17-driven autoimmunity
IL-10	↓ (or dysregulated)	Loss of anti-inflammatory control

Elevated levels of these cytokines have been observed in blood, saliva, and cerebrospinal fluid of PTSD patients, and are associated with disease activity in autoimmune conditions^{[18][12][16]}.

Immune Cell Phenotypes and Adaptive Immunity Changes

Chronic stress and trauma alter the balance of immune cell subsets. There is an increase in pro inflammatory Th17 cells (producing IL-17) and a reduction in regulatory T cells (Tregs, producing

IL-10 and TGF- β), tipping the immune system toward autoimmunity^{[7][13]}. Macrophages and dendritic cells also become more pro-inflammatory, and natural killer (NK) cell activity is reduced^[10].

Table: Immune Cell Changes in Chronic Stress/PTSD

Immune Cell Type	Change in Chronic Stress/PTSD	Functional Consequence
Th17 Cells	↑ IL-17 production	Drives autoimmunity (RA, MS, IBD)
Treg Cells	↓ IL-10, ↓ function	Loss of immune tolerance
Macrophages	↑ TNF- α , IL-1 β , IL-6	Sustained inflammation
NK Cells	↓ Activity	Impaired immune surveillance

The reciprocal regulation of Th17 and Treg cells is critical; factors such as IL-6 and IL-23 promote Th17 differentiation, while TGF- β and IL-2 support Treg development. Dysregulation of this balance is a hallmark of autoimmunity^[7].

Epigenetic and Gene-Expression Changes After Trauma

Trauma and chronic stress induce epigenetic modifications, particularly DNA methylation changes in genes regulating immune and stress responses^{[13][12]}. For example, altered methylation of the NR3C1 gene (encoding the glucocorticoid receptor) and immune-related genes such as AIM2 and IL-18 has been linked to PTSD severity and increased inflammatory markers (e.g., CRP)^[12]. These epigenetic changes can persist for years, explaining the long latency between trauma exposure and autoimmune disease onset^[11].

Gut Microbiome, Gut Permeability, and the Gut-Brain-Immune Axis

The gut microbiome is a key regulator of immune homeostasis. Trauma and chronic stress disrupt gut microbial diversity and composition, leading to increased gut permeability ("leaky gut") and systemic inflammation^[7]. The gut-brain-immune axis involves bidirectional communication via neural (vagus nerve), hormonal, and immune pathways. Dysbiosis and increased gut permeability allow microbial products to enter circulation, activating immune responses and promoting autoimmunity^[7].

Animal models demonstrate that stress-induced changes in the microbiome can alter behavior, immune responses, and susceptibility to autoimmune diseases. Human studies have found that PTSD is associated with reduced microbial diversity and increased abundance of pro inflammatory bacteria^[15].

Neuroimmune Interactions and Microglial Activation

Chronic stress and trauma activate microglia, the resident immune cells of the central nervous system, leading to neuroinflammation, synaptic pruning, and cognitive dysfunction^{[13][9]}. Pro inflammatory cytokines produced by microglia and infiltrating immune cells can cross the blood brain barrier, further amplifying neuroimmune activation and contributing to both psychiatric

and autoimmune pathology^[13].

Disease-Specific Associations

Rheumatoid Arthritis (RA)

Multiple cohort studies and meta-analyses have established a significant association between PTSD/chronic stress and RA. In the Nurses' Health Study II, women with ≥ 4 PTSD symptoms had a 76% increased risk of developing RA (HR = 1.76, 95% CI: 1.16-2.67), independent of smoking and other confounders^[1]. Animal models of arthritis confirm that stress exacerbates disease severity, increases pro-inflammatory cytokines, and alters immune cell function^[14].

Systemic Lupus Erythematosus (SLE)

PTSD is strongly associated with increased risk of SLE. A systematic review found that PTSD patients had at least a twofold increased risk of developing SLE, with some studies reporting nearly threefold risk, especially among women^[1]. The risk is higher in those with greater trauma exposure and PTSD severity^[3].

Multiple Sclerosis (MS)

PTSD and chronic stress are linked to increased risk and greater disease activity in MS. Veterans with MS and comorbid PTSD had higher relapse rates, more MRI lesions, and faster disability progression than controls^[1]. The risk ratio for MS in PTSD patients is approximately 1.3 (95% CI: 1.04-1.64)^[1].

Inflammatory Bowel Disease (IBD)

PTSD is associated with increased risk of IBD, including Crohn's disease and ulcerative colitis. The risk ratio for IBD in PTSD patients is 1.14 (95% CI: 1.09-1.20)^[1]. PTSD is also bi-directionally correlated with GI diseases such as IBS, GERD, and functional dyspepsia, with particularly strong associations for IBS (odds ratio >2)^[15].

Autoimmune Thyroid Disease

The relationship between PTSD and autoimmune thyroid diseases is complex. Mendelian randomization studies suggest a potential increase in the risk of Graves' disease (GD) associated with PTSD (OR = 1.056, 95% CI: 1.008-1.105), but no significant association with autoimmune thyroiditis (AIT)^[4]. Epidemiological studies support a link between stress and GD, but not AIT, possibly due to differences in Th1/Th2 immune responses^[4].

Table: Disease-Specific Associations Between Trauma/PTSD and Autoimmune Diseases

Autoimmune Disease	Risk Ratio/HR (PTSD vs. Controls)	Key Findings/Notes
Rheumatoid Arthritis (RA)	HR 1.4-1.8	Dose-response with PTSD symptom severity
Systemic Lupus Erythematosus (SLE)	≥ 2.0 (up to 3.0)	Strongest in women,

		dose response
Multiple Sclerosis (MS)	RR 1.3	Higher relapse, MRI activity, disability
Inflammatory Bowel Disease (IBD)	RR 1.14	Bi-directional, especially with IBS
Graves' Disease (GD)	OR 1.06	Positive association, Th2-driven
Autoimmune Thyroiditis	No significant association	Th1-driven, not linked to PTSD

Psychological Factors: Unresolved Trauma, Coping Styles, and Chronic Stress Responses

Unresolved trauma perpetuates chronic stress responses, maintaining the body in a state of hyperarousal, fear, and immune activation^[8]. Maladaptive coping styles, such as avoidance or rumination, exacerbate stress and inflammation. Early-life adversity, including ACEs, has a particularly strong and lasting impact on immune regulation, increasing the risk of autoimmune disease in adulthood^[3].

Psychological resilience and adaptive coping can mitigate the adverse effects of trauma and stress on immune function. Individuals with higher resilience exhibit more favorable cytokine profiles and reduced inflammation^[12]. Interventions that address trauma, promote emotional processing, and foster adaptive coping can modulate inflammatory pathways and may reduce autoimmune risk^[16].

Interventions: Trauma-Focused Therapy, Stress Reduction, and Effects on Inflammation/Autoimmunity

Psychotherapy and Mind-Body Interventions

Trauma-focused therapies, such as cognitive behavioral therapy (CBT), eye movement desensitization and reprocessing (EMDR), and mindfulness-based stress reduction (MBSR), have demonstrated efficacy in reducing PTSD symptoms and modulating inflammatory markers^[16]. For example, yoga interventions in women veterans with PTSD led to increases in anti-inflammatory cytokines (IL-10) and improvements in psychological well-being^[16].

Lifestyle Modifications

Lifestyle adjustments, including regular physical activity, restorative sleep, anti-inflammatory diets, and social support, can reduce stress and inflammation, supporting immune balance^[7]. Practices that enhance vagal tone, such as meditation and deep breathing, may improve neuroimmune regulation and reduce autoimmune flares^[7].

Pharmacological Interventions

Preliminary evidence suggests that selective serotonin reuptake inhibitors (SSRIs) and glucocorticoid therapy in the acute phase after trauma may attenuate the risk of developing autoimmune disease^[9]. However, further research is needed to clarify the efficacy and safety of these approaches in patients with comorbid PTSD and autoimmunity.

Multidisciplinary Care

Expert consensus emphasizes the importance of multidisciplinary care for patients with autoimmune diseases and comorbid trauma or stress disorders. This includes patient education, psychosocial support, physical therapy, and coordinated medical and mental health care^[7].

Clinical Implications: Screening, Monitoring, and Multidisciplinary Care

Given the robust association between trauma, chronic stress, and autoimmunity, clinicians should routinely screen for trauma histories and PTSD symptoms in patients presenting with autoimmune diseases, and vice versa^[4]. Early identification and intervention may mitigate disease risk and improve outcomes. Monitoring inflammatory markers (e.g., CRP, IL-6) and immune cell phenotypes can provide insight into disease activity and guide treatment decisions^[16].

Multidisciplinary care teams, including rheumatologists, psychiatrists, psychologists, and primary care providers, are essential for addressing the complex interplay of psychological and physical health in these patients^[7].

Methodological Issues, Confounders, and Causality Assessment

While the association between trauma, chronic stress, and autoimmunity is robust, several methodological challenges remain. These include:

- **Confounding factors:** Smoking, alcohol use, and other lifestyle factors may contribute to both PTSD and autoimmune risk, but most studies have adjusted for these variables and found little impact on effect estimates^[5].
- **Diagnostic lag:** The long latency between trauma exposure and autoimmune disease onset complicates temporality assessment^[1].
- **Surveillance bias:** Increased healthcare utilization by PTSD patients may lead to higher detection rates of autoimmune diseases^[1].
- **Measurement variability:** Differences in PTSD severity assessment and autoimmune disease classification may affect risk estimates^[1].

Despite these limitations, the consistency of findings across diverse populations and

study designs supports a causal relationship^[1].

Sex Differences, Age, and Demographic Moderators

Women have higher rates of both PTSD and autoimmune diseases, but the magnitude of the trauma-autoimmunity association is similar in men and women^[1]. Age at trauma exposure is a critical moderator; younger individuals and those with early-life adversity are at greater risk for autoimmune disease^[3]. Sex hormones, such as estrogen, may modulate HPA axis sensitivity and immune responses, contributing to sex differences in disease susceptibility^[21].

Epistemic Synthesis: Summary Tables

Table: Biological Mechanisms Linking Trauma/Stress to Immune Dysregulation

Mechanism	Description
HPA Axis Dysregulation	Impaired cortisol signaling, glucocorticoid resistance, loss of anti-inflammatory control
SNS Activation	Chronic catecholamine release, altered immune cell function, increased inflammation
Cytokine Imbalance	↑ IL-6, TNF-α, IL-1β, IL-17, CRP; ↓ IL-10, TGF-β
Th17/Treg Imbalance	↑ Th17 (pro-inflammatory), ↓ Treg (regulatory), loss of tolerance
Epigenetic Modifications	DNA methylation changes in immune and stress response genes
Gut-Brain-Immune Axis Disruption	Microbiome dysbiosis, increased gut permeability, systemic inflammation
Neuroimmune Activation	Microglial activation, neuroinflammation, cognitive dysfunction

Table: Autoimmune Diseases Associated with Trauma/Chronic Stress

Disease	Strength of Association (RR/HR)	Notes
Rheumatoid Arthritis (RA)	1.4-1.8	Dose-response with PTSD symptom severity
Systemic Lupus Erythematosus	≥2.0 (up to 3.0)	Strongest in women, dose response

Multiple Sclerosis (MS)	1.3	Higher relapse, MRI activity, disability
Inflammatory Bowel Disease (IBD)	1.14	Bi-directional, especially with IBS
Graves' Disease (GD)	1.06	Positive association, Th2-driven
Autoimmune Thyroiditis	No significant association	Th1-driven, not linked to PTSD

Table: Key Studies Supporting the Trauma-Autoimmunity Link

Study/Population	Autoimmune Diseases	Key Findings (Risk Ratios/HRs)
Millennium Cohort (US Military)	RA, SLE, MS, IBD	HR = 1.6 for any autoimmune disease with PTSD
Nurses' Health Study II (Women)	RA, SLE, Thyroiditis	HR = 1.76 for RA with ≥ 4 PTSD symptoms
Swedish National Registers	41 autoimmune diseases	RR = 1.09-1.49 (varies by disease)
ACE Study (US Adults)	21 autoimmune diseases	70-100% increased risk with ≥ 2 ACEs
Veterans Health Administration	RA, SLE, MS, IBD, Thyroiditis	Elevated risks in PTSD vs. controls
Meta-analysis (8 studies, n~ 2M)	IBD, SLE, RA, MS, Thyroiditis	RR = 1.14-1.30 (disease-specific)

Research Gaps, Priorities, and Future Directions

Despite significant advances, several research gaps remain:

- **Mechanistic clarity:** The precise molecular pathways linking chronic stress to the transition from immune tolerance to autoimmunity require further elucidation^[7].
- **Reversibility:** The potential for reversing HPA axis alterations and immune dysfunction after trauma resolution is not fully understood^[7].
- **Intervention efficacy:** More randomized controlled trials are needed to assess the impact of trauma-focused therapies and stress reduction on autoimmune outcomes^[16].
- **Demographic moderators:** The roles of sex, age, and genetic background in moderating risk and resilience need further study^[21].

- **Multi-omics approaches:** Integrating genomics, epigenomics, transcriptomics, and microbiomics will enhance mechanistic understanding and identify novel therapeutic targets [11].

Conclusion

The correlation between unprocessed emotional trauma, chronic stress, inflammation, and the development of autoimmune diseases is supported by robust epidemiological evidence and a growing understanding of the underlying biological mechanisms. Chronic stress and unresolved trauma disrupt neuroendocrine and immune regulation, fostering a pro-inflammatory

environment, altering immune cell phenotypes, and promoting loss of tolerance to self antigens. These changes increase susceptibility to a range of autoimmune diseases, including RA, SLE, MS, IBD, and Graves' disease. Early-life adversity and maladaptive coping further amplify risk, while psychological resilience and targeted interventions can mitigate adverse outcomes. Multidisciplinary care, routine screening, and integrated research efforts are essential for addressing the complex interplay of psychological and physical health in trauma exposed populations. Future research should prioritize mechanistic studies, intervention trials, and longitudinal analyses to optimize prevention and treatment strategies for stress-related immune dysregulation and autoimmunity.

References (21)

1. *Posttraumatic stress disorder and risk of selected autoimmune diseases*
<https://link.springer.com/article/10.1186/s12888-020-2432-9>
2. *Causal effects of post-traumatic stress disorder on autoimmune thyroid*
<https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2024.1417302/full>
3. *Systematic review and meta-analysis of post-traumatic stress disorder*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11882857/>
4. *Can Trauma Cause Autoimmune Disease? The Evidence.* <https://scienceinsights.org/can-trauma-cause-autoimmune-disease-the-evidence/>
5. *Post-traumatic stress disorder and systemic lupus erythematosus*
<https://pubmed.ncbi.nlm.nih.gov/40257431/>
6. *The long-term costs of traumatic stress: intertwined physical and*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2816923/>
7. *The Balance of Th17 versus Treg Cells in Autoimmunity - PMC.*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5877591/>
8. *Inflammatory and oxidative stress markers in post-traumatic stress*
<https://www.nature.com/articles/s41380-022-01564-0>
9. *Research Progress on Inflammation and Immune Dysregulation in PTSD.*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12171638/>

10. *The Emerging Role of the Gut Microbiome in Posttraumatic Stress*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10591863/>
11. *Effects of group trauma-sensitive yoga on inflammatory markers and*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12981235/>
12. *Impact of comorbid post traumatic stress disorder on multiple sclerosis*
<https://pubmed.ncbi.nlm.nih.gov/34854320/>
13. *Chronic Stress and Autoimmunity: The Role of HPA Axis and ... - MDPI.*
<https://www.mdpi.com/1422-0067/26/20/9994>
14. *The sympathetic nervous response in inflammation - PMC.*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4396833/>
15. *Neuroinflammation in Post-Traumatic Stress Disorder - PMC.*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9138406/>
16. *DNA Methylation Correlates of PTSD: Recent Findings and Technical*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC6314898/>
17. *The Relationship Between Inflammation and Post-traumatic Stress*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8385235/>
18. *Posttraumatic Stress Disorder and Risk for Incident Rheumatoid Arthritis.* <https://pmc.ncbi.nlm.nih.gov/articles/PMC4740283/>
19. *Sex Differences in Post-Traumatic Stress Disorder Risk: Autonomic*
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7598146/>